MIL-L-46959B (AR)
8 February 1980
SUPERSEDING
MIL-L-46959A (MU)
3 May 1965

MILITARY SPECIFICATION

LUCKY ASSEMBLY, INSULATED, FOR CARTRIDGES 105MM, HEAT-T, M456A1 AND 90MM, HEAT-T, M431A2 PARTS FOR, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the manufacture of parts for, assembling and packing for one type of insulated lucky assembly for use in Crtgs, 105MM, HEAT-T, M456A1 and 90MM, HEAT-T, M431A2

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-1167 - Ammunition Data Card
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

8861458 - Insulated Lucky Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assemblies.-The assemblies and all components shall comply with all requirements specified on Dwg. 8861458 and other applicable drawings and specifications.

3.3 Activity.-The lucky shall have a piezo electric coefficient, d_{33}, of not less than 9×10^{-10} coulombs per pound where d_{33} relates the charge to the load and is measured along the axis of polarization. The piezo-electric coefficient is determined by measuring the charge developed while releasing the lucky from a compressive pressure 1000 pound per square inch (p.s.i.) for units of .012 to .500 square inch electrode area and 500 p.s.i. for .501 and 1.000 square inch areas.

3.4 Dielectric constant.-The lucky shall exhibit a dielectric constant of not less than 2000 relative to air.

3.5 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

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3.6 Workmanship.-All parts shall be of the proper finish, dry and free of burrs, cracks, chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.7 Shear test.-Shear strength of bonded components of the lucky assembly shall be 50 pounds, minimum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 10 each components and assemblies which have been manufactured, assembled and packed using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed.-See MIL-A-48078. All First Article Samples shall be subjected to all inspections/tests specified in tables beginning with Paragraph 4.4.2.1.

4.3.3 Rejection.-See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-46959B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8861459
4.4.2.1	Plug				NEXT HIGHER ASSEMBLY 8861458
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of stem, min.		0.40%	3.2	Gage
102	Flatness		0.40%	3.2	Gage
103	Protective finish missing, improper of damaged		0.40%	3.2	Visual
104	Surface finish improper		0.40%	3.2	Visual
Minor					
201	Diameter of stem, max.		0.65%	3.2	Gage
202	Runout		0.65%	3.2	Gage
203	Perpendicularity		0.65%	3.2	Gage
204	Diameter of flange		0.65%	3.2	Gage
205	Thickness of flange		0.65%	3.2	Gage
206	Total length		0.65%	3.2	Gage
207	Radius missing from end of stem		0.65%	3.2	Visual
208	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-46959B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Disc				8861461
					NEXT HIGHER ASSEMBLY
					8861458
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Protective finish missing, improper or damaged		0.40%	3.2	Visual
102	Surface finish		0.40%	3.2	Visual
Minor					
201	Diameter		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Evidence of poor workmanship, including chips and burrs		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-L-46959B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Lucky Assembly	1 OF 1	8864161
			NEXT HIGHER ASSEMBLY 8861458

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Activity		100%	3.3	Test/4.5.1
102	Dielectric constant		100%	3.4	Test/4.5.2
103	Distance from plug to edge of lucky, min.		0.40%	3.2	Gage
104	Distance from disc to edge of lucky, min.		0.40%	3.2	Gage
105	Cement on edge of lucky		0.40%	3.2	Visual
106	True position		0.40%	3.2	Gage
107	Foreign matter on any components prior to bonding		0.40%	3.2	Visual
108	Cement or bonding process improper		0.40%	3.2	Visual
109	Shear test	10		3.7	Test/4.5.3
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-46959B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Insulator				8861462
					NEXT HIGHER ASSEMBLY
					8861458
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material defective or damaged		0.40%	3.2	Visual
Minor					
201	Presence of flash		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-46959B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Insulated Lucky Assembly				8861458
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component(s) missing or damaged		0.40%	3.2	Visual
102	Improper assembly		0.40%	3.2	Visual
103	Any loose component(s)		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Activity.-In the event that the inspector discovers that the test has not been fully complied with, or that non-conforming units have not been rejected, he shall reject all units tested since his last satisfactory surveillance. The contractor shall retest these rejected units 100% in the presence of the inspector. Units failing to meet the requirements during retest shall be classified defective and removed from the lot.

4.5 Methods of inspection

4.5.1 Activity.-After polarization, the lucky shall be heated to a temperature of 165 degrees Fahrenheit (^{O}F) + $10^{O}F$ and maintained at that temperature for a period of 10 minutes. They shall then be cooled to $75^{O} \pm 10^{O}F$ and maintained at that temperature for not less than 24 hours and shall then be tested to determine conformance to this specification.

4.5.2 Determination of dielectric constant.-After performing the activity specified in 4.5.1, the dielectric constant shall be determined using the following formula:

$$K = \frac{4.44\ Ct}{A}$$

Where:

K = dielectric constant of lucky
C*= capacitance of lucky in micro-micro-farads
t = maximum thickness of lucky (inches)
A = minimum area of lucky electrode surface (square inches)

*Capacitance shall be determined using a bridge or meter which applies a measuring signal of 1,000 cycles per second. The test shall be conducted at a temperature between $+65^{O}F$ and $+85^{O}F$.

4.5.3 Shear test.-The lucky assembly shall be mounted in a suitable test fixture and a lateral load shall be applied at the rate of .100 to .125 inch/minute. Should any break-away force value of any of the samples tested not meet the minimum shear requirement, the lot represented by the sample shall be rejected.

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5. PACKAGING

5.1 Preservation and packaging

5.1.1 Level B.-Insulated lucky assemblies shall be preserved, packaged and marked in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Intended use.-Insulated lucky assemblies covered by this specification are intended for use on cartridges, 105MM, HEAT-T, M456A1 and 90MM, HEAT-T, M431A2.

6.2 Ordering data.-See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals.-See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Drawings.-Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Submission of ammunition data cards.-In addition to the normal distribution of records when the assembly is procured by the Army Armament Materiel Readiness Command, one copy of all ammunition data cards shall be forwarded to each of the following: Commander, ARRADCOM, ATTN: DRDAR-LCU-CV and DRDAR-QAR-E, Dover, NJ 07801; and Commander, ARRCOM, ATTN: DRSAR-QAD, Rock Island, IL 61299.

6.6 It is noted that the size of inspection lots of components, assemblies, or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items for delivery may be equivalent to such contract quantities provided the lot formation criteria and sampling provisions of this specification are maintained.

Custodian:
ARMY-AR

Preparing Activity:
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